Backend

- Internet
  - How does the internet work?
  - What is HTTP?
  - What is Domain Name?
  - What is hosting?
  - DNS and how it works?
  - Browsers and how they work?
- Pick a Language
  - JavaScript
  - Go
  - Python
  - Ruby
  - Java
  - C#
  - PHP
  - Rust
- Version Control Systems
  - Git
  - Repo Hosting Services
    - GitHub
    - GitLab
    - Bitbucket
- Relational Databases
  - PostgreSQL
  - MySQL
  - MariaDB
  - MS SQL
  - Oracle
  - SQLite
- Learn about APIs
  - HATEOAS
  - Open API Specs
  - REST
  - JSON APIs
  - SOAP
  - gRPC
  - GraphQL
  - Authentication
    - JWT
    - OAuth
    - Basic Authentication
    - Token Authentication
    - Cookie Based Auth
    - OpenID
    - SAML
- Caching
  - Redis
  - Memcached
  - Server Side
  - CDN
  - Client Side
- Testing
  - Integration Testing
  - Unit Testing
  - Functional Testing
- CI / CD
- Web Security
  - MD5
  - SHA
  - scrypt
  - bcrypt
  - Hashing Algorithms
  - HTTPS
  - OWASP Risks
  - CORS
  - SSL/TLS
  - CSP
  - Server Security
  - API Security Best Practices
- More about Databases
  - ORMs
  - ACID
  - Transactions
  - N+1 Problem
  - Normalization
  - Failure Modes
  - Profiling Perfor.
  - Migrations
- Scaling Databases
  - Database Indexes
  - Data Replication
  - Sharding Strategies
  - CAP Theorem
- Software Design & Architecture
- Architectural Patterns
  - Monolithic Apps
  - Microservices
  - SOA
  - Serverless
  - Service Mesh
  - Twelve Factor Apps
- Design and Development Principles
  - GOF Design Patterns
  - Domain Driven Design
  - Test Driven Development
  - CQRS
  - Event Sourcing
- Containerization vs Virtualization
  - Docker
  - LXC
  - Kubernetes
- Web Servers
  - Nginx
  - Apache
  - Caddy
  - MS IIS
- Search Engines
  - Elasticsearch
  - Solr
- Message Brokers
  - RabbitMQ
  - Kafka
- Real-Time Data
  - Server Sent Events
  - WebSockets
  - Long Polling
  - Short Polling
  - GraphQL
- NoSQL Databases
  - Document DBs
    - MongoDB
    - CouchDB
  - Key-Value
    - Redis
    - DynamoDB
  - Realtime
    - Firebase
    - RethinkDB
  - Time Series
    - Influx DB
    - TimeScale
  - Column DBs
    - Cassandra
    - Base
  - Graph DBs
    - Neo4j
    - AWS Neptune
- Building For Scale
  - Graceful Degradation
  - Throttling
  - Backpressure
  - Loadshifting
  - Circuit Breaker
  - Mitigation Strategies
  - Migration Strategies
  - Types of Scaling
  - Metrics logging and other observable items that can help in debugging and solving Issues when things go wrong.
    - Observability
  - Difference & Usage
    - Instrumentation
    - Monitoring
    - Telemetry
- Visit the DevOps Beginner Roadmap
  - Basic Infrastructure Knowledge